Quantification: 543_3
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	11
interior lacuna count	10
frame-edge lacuna count	1

Interior lacunae only: mean, median and sample SD (n = 10)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2218.44	2174	555.066	px^2
major_axis_length_px	lacuna shape	81.5294	80.8669	9.8489	px
minor_axis_length_px	lacuna shape	37.3118	35.2412	10.5528	px
aspect_ratio	lacuna shape	2.2853	2.2947	0.5009	unitless
eccentricity	lacuna shape	0.8782	0.9	0.0743	unitless
solidity	lacuna shape	0.8919	0.9295	0.0895	unitless
roots_count	headline	7.4	7	3.5024	count
ring_length_r30_px	headline	393.1	415	141.427	px
ring_length_r60_px	reported	1014.8	1102.5	352.584	px
owned_length_px	ownership-dependent	3708.2	3425.46	1685.36	px
edge_count	ownership-dependent	172.9	157	82.7532	count
mean_edge_length_px	ownership-dependent	21.762	21.4494	2.4751	px
perimeter_px	normalised	216.655	219.693	74.8607	px
ring_area_r30_px2	normalised	8553.6	8576.5	1974.23	px^2
ring_area_r60_px2	normalised	21737.1	21875	4704.23	px^2
in_frame_fraction_r30	normalised	0.9906	1	0.0298	unitless
in_frame_fraction_r60	normalised	0.969	1	0.0853	unitless
ring_density_r30	normalised	0.0462417	0.0483211	0.0130488	px^-1
ring_density_r60	normalised	0.0469657	0.0499945	0.0130409	px^-1
roots_per_100px_perimeter	normalised	3.6309	3.5922	1.5853	per 100 px
ring_attached_length_r30_px	network v2	366	389	137.712	px
ring_attached_length_r60_px	network v2	925.4	989.5	336.241	px
ring_length_w_r30_px	network v2	447.389	468.549	164.391	px
ring_length_w_r60_px	network v2	1158.13	1248.59	404.414	px
sholl_crossings_r10	network v2	10.3	10.5	4.1913	count
sholl_crossings_r20	network v2	12.7	14	4.5228	count
sholl_crossings_r30	network v2	15	15.5	5.0111	count
loop_count	loops, report only	21	17.5	15.0185	count
loop_edge_share	loops, report only	0.4174	0.4462	0.1447	share
ring_width_mean_r30_px	width	7.32	7.1581	0.6533	px

Per field
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measure	value	units
analysed_area_px2	1.02792e+06	px^2
total_skeleton_length_px	38903	px
canalicular_length_density_per_px	0.0378464	px^-1
junction_count	1012	count
junction_density_per_px2	0.00098451	px^-2
lacuna_count	11	count

lacunae_per_px2	1.07e-05	px^-2
skeleton_component_count	155	count
mean_component_length_px	250.987	px
median_component_length_px	19	px
field_length_density_w_per_px	0.0431121	px^-1
field_density_without_flagged_per_px	0.0378464	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	211	px
unattached_component_count	14	count
canalicular_length_density_attached_per_px	0.0378464	px^-1
field_length_density_with_unattached_per_px	0.0380517	px^-1
excluded_area_px2	0	px^2
analysed_field_area_px2	1.04858e+06	px^2
loop_count	243	count
loop_edge_share	0.4656	share
edge_count	2036	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.839	px
width_median_px	8	px
width_p10_px	4.4721	px
width_p90_px	10.198	px
width_pixels_used	38903	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	2174	80.8669	35.2412	2.2947	0.9	0.9518	0.4109	79.3965	270.898	False
2	3447	90.3677	63.9486	1.4131	0.7066	0.674	0.4223	197.059	916.902	False
3	1883	69.2617	34.8694	1.9863	0.864	0.9448	0.5742	250.833	340.47	False
4	2052	69.2015	39.4228	1.7554	0.8219	0.9112	0.0622	388.671	369.939	False
5	2266	88.1613	35.5691	2.4786	0.915	0.9021	0.1324	473.05	745.191	False
6	1625	80.1248	27.6985	2.8927	0.9383	0.8325	0.2321	545.572	877.076	False
7	1683	71.0123	31.3735	2.2634	0.8971	0.9355	-0.0518	634.68	198.315	False
8	2203	91.2092	31.4053	2.9043	0.9389	0.9295	0.0832	734.301	635.44	False
9	2633	93.5592	36.2778	2.579	0.9218	0.9454	0.0457	780.576	481.495	False
10	531	28.7881	24.8252	1.1596	0.5063	0.9332	0.5942	1012.28	184.834	True
11								211.745	24.6149	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	7	416	1138	2105.38	94	22.3977	7.0372
2	14	628	1492	4480.64	214	20.9376	7.0243
3	8	414	1067	6578.32	308	21.3582	8.3894
4	8	402	998	3812.68	177	21.5406	7.6629
5	12	425	1154	3038.25	132	23.017	8.1921
6	6	523	1388	5808.94	298	19.4931	6.3897

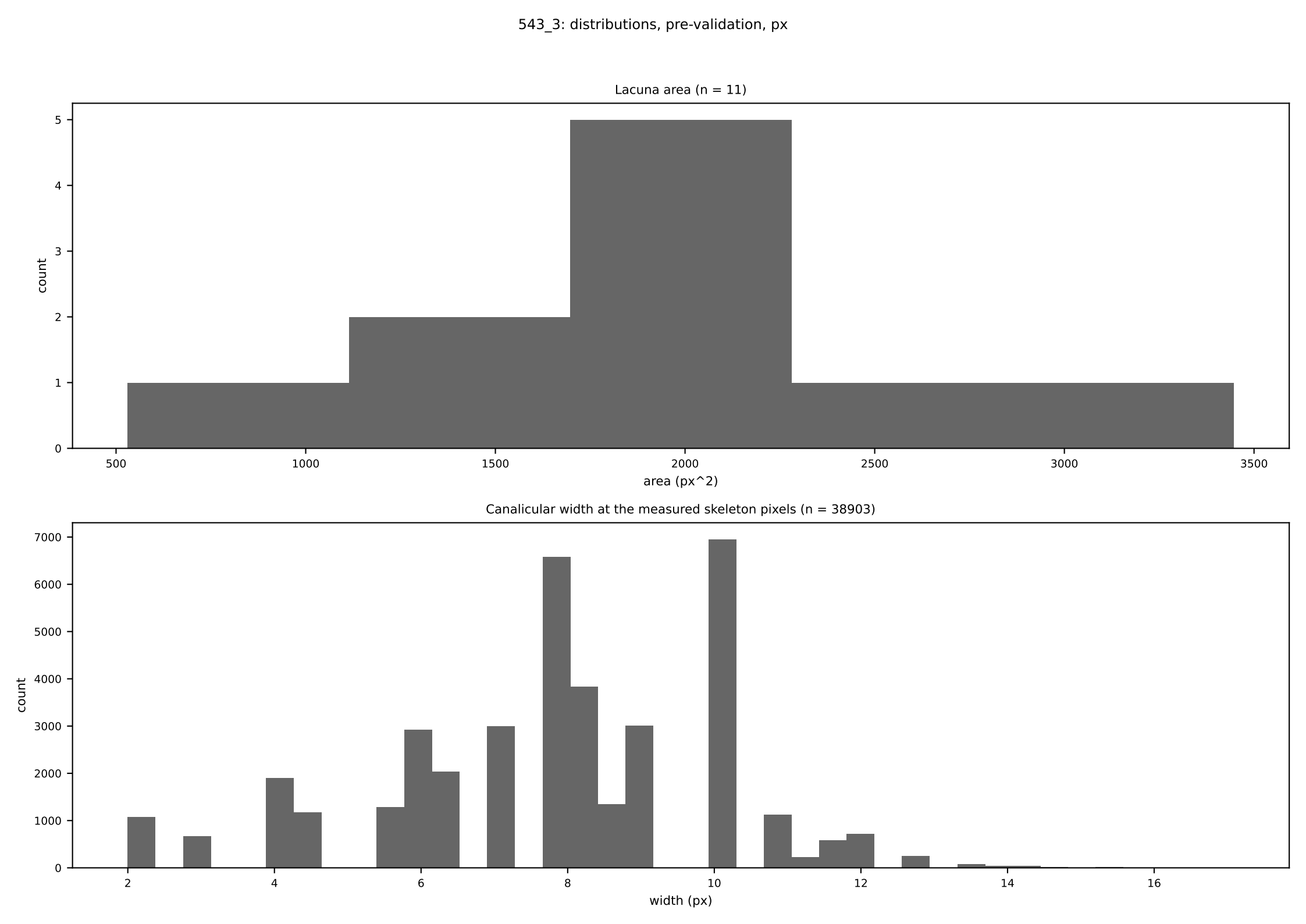
7	2	116	325	2775.38	137	20.2582	6.7134
8	6	432	1140	4837.25	201	24.0659	7.7086
9	7	339	885	1948.56	73	26.6926	7.2789
10	3	188	404	1228.23	62	19.8102	5.6218
11	4	236	561	1696.59	95	17.8589	6.8039

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	210.066	8595	21715	1	0.9616	0.0484002	0.0524062	3.3323
2	363.446	12097	29071	1	1	0.0519137	0.0513226	3.852
3	184.794	7941	19515	1	1	0.0521345	0.0546759	4.3291
4	205.965	8333	20507	1	1	0.0482419	0.0486663	3.8841
5	241.907	9380	23807	1	1	0.0453092	0.0484731	4.9606
6	229.321	8558	22035	1	1	0.0611124	0.0629907	2.6164
7	191.965	7943	21525	1	1	0.0146041	0.0150987	1.0419
8	238.108	9212	24077	1	1	0.0468954	0.0473481	2.5199
9	242.936	9386	24420	1	1	0.0361176	0.0362408	2.8814
10	95.3909	3354	9525	0.6102	0.5725	0.0560525	0.0424147	3.145
11	58.0416	4091	10699	0.9057	0.7287	0.0576876	0.0524348	6.8916

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	386	984	481.517	1327.77	9	15	17
2	603	1399	726.683	1702.55	19	17	19
3	392	1049	459.321	1214.25	10	13	18
4	382	945	456.676	1155.63	11	15	15
5	425	995	477.777	1308.53	11	14	14
6	466	1321	602.801	1582.25	12	16	22
7	108	309	132.983	363.451	4	3	4
8	413	1051	489.676	1282.93	13	14	16
9	257	655	383.534	1006.34	9	14	15
10	172	352	215.167	456.191	5	7	6
11	228	546	262.924	637.558	5	6	10


543_3: distributions, pre-validation, px
Lacuna area (n = 11)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 38903)
count
width (px)